UI and UX Enhancement for the Hyperspectral Coffee and Cocoa Monitoring Web App

Context

We currently have a working Python prototype for hyperspectral coffee and cocoa detection and health assessment using Earth Engine, EMIT Level 2A reflectance, Sentinel 2, and an interactive geospatial workflow. The analytical backend is already functional, but the current interface is still prototype level and needs improvement in usability, visual clarity, workflow design, and readiness for deployment in a Hugging Face style web application.

Objective

Design and implement a more user friendly and app ready interface for the existing workflow. The goal is to improve the usability of the current web application, strengthen the map based interaction model, and make the experience clearer for non technical end users while preserving the current analytical logic.

Scope of Work

The applicant should focus on the frontend interaction flow, geospatial visualization quality, and UI structure rather than changing the core remote sensing methodology.

Expected Functional Improvements

1. **Workflow redesign**

Redesign the current user flow so the main steps are clearly structured and easy to follow. At minimum, the interface should guide the user through:

- AOI selection
- data loading
- crop detection
- health assessment
- map filtering and output interpretation

The final interface should reduce confusion and improve task completion without requiring the user to understand the underlying code.

2. **UI refinement**

Improve the visual layout and interaction design of the app. This can include:

- clearer button grouping
- better spacing and hierarchy
- more intuitive labels
- step indicators or progress guidance
- consistent typography and color use
- improved error and status messages

3. **Map visualization enhancement**

Improve the way results are shown on the geospatial interface. This can include:

- clearer legends
- more intuitive color palettes for crop classes and health status
- better layer visibility control
- stronger visual distinction between coffee cocoa and background
- improved AOI highlighting
- more interpretable plot boundaries and health overlays

4. **Filtering and interaction usability**
Improve the filtering system so that users can easily:

- show only coffee
- show only cocoa
- display selected health categories
- understand what is currently active on the map

The interaction should be as simple and direct as possible.
5. App readiness and modularity
Refactor the current app code into cleaner modules or components so it is easier to integrate into the Hugging Face application stack. The code should become easier to maintain, extend, and deploy.
Optional Stretch Feature
If time permits, implement one additional UX feature that meaningfully improves usability, for example:

- collapsible side panel
- reset and undo interaction improvements
- dynamic legend
- AOI preset cards
- better loading indicators
- downloadable result summary section

Constraints

- Use the current Python based application context and existing code as the starting point
- Do not redesign the analytical methodology unless strictly necessary for UI stability
- Focus on web application interface, interaction quality, and usability
- Keep the output compatible with potential integration into the current Hugging Face app

Deliverables
1. **Code contribution**
Provide revised application code or UI module improvements that can be reviewed and potentially integrated into the current app.
2. **Short written report, 1 to 2 pages**
The report should explain:

- what UI and UX changes were made
- why those changes improve usability
- how the revised design supports map based interaction
- what parts are immediately reusable in the current app
- what would be the next logical improvement

- 3. **Visual evidence**

Provide screenshots or a short screen recording demonstrating:

- improved workflow
- map filtering
- crop visualization
- health status visualization
- any additional UX enhancements

Evaluation Criteria
The submission will be evaluated based on:

- clarity and usability of interface design
- strength of map visualization and geospatial interaction
- code organization and integration readiness
- practical UX judgment
- relevance of improvements to the existing application
- overall polish and usability of the web app experience

Suggested Workload
Estimated effort is approximately 6 to 10 hours of focused work.
Short assignment version you can send
Please improve the current hyperspectral coffee and cocoa monitoring app by redesigning the user interface and interaction flow, enhancing the geospatial visualization, and refactoring the code into a cleaner app ready structure. The goal is to make the workflow more intuitive, more visually interpretable, and easier to integrate into the Hugging Face app while preserving the existing analytical logic.